

Research in the field of sensors

Research is an ongoing process that is not limited to any product or any component. Each year, the research process gets enhanced. Sensors are the products that have gained importance due to the type of research incorporated in this field. Researchers are trying to find ways to decrease the size of sensors and to make their use in almost all sectors, from farming to aerospace and from consumer electronics to medical equipment. Latest concepts of some research done in 2013 are as follows:

1. A cellphone-like sensor to help paralysed patients control things with their thoughts. These sensors help control the things by relaying real-time broadband signals from up to 100 neurons in freely moving objects inductively. They transmit data at 24Mbps via 3.2GHz and 3.8GHz microwave frequencies to an external receiver. Once being charged for two hours, they can operate wirelessly for more than six hours.
2. Smart dust are the energy autonomous sensor networks that can be sprinkled over large farmlands to measure details such as level of sunlight and moisture in the soil and to relay the same data to a central interface. This data would help the farmers to regulate the amount of watering and even to predict the quality and type of crop yield.
3. Freight intelligent detection and contingency command system that uses sensors for cargo identification, cargo state detection, cargo public data services and cargo contingency command to control accidents and prevent destruction of power. This technology was used before with the help of RFID. However, areas or regions with no infrastructure to support tracking cannot use RFID. In such places, sensors built on a chip help in cellular tracking. The real-time sensors that are used in this technology help reduce spoilage and reduce theft allowing speedy recovery of goods.
4. Multimodal bio-image sensor that has been devised for filterless fluorescence imaging and proton concentration (pH) imaging. The prototype of this sensor has been devised using silicon-integrated CMOS circuit technology. Using this sensor fluorescence and pH, both were obtained successfully in the same area in real time and that too without the use of any optical filters or gratings. This sensor uses a pH sensor, which in turn uses a charge-coupled device image sensor technology. This technology is combined with a fluorescence filterless detector in the same pixel. This type of sensor can be used for intra- and extracellular recording.
5. 'Matchstick' is a room-to-room thermal model that is used to predict the indoor temperature from wireless sensor data. The data is gathered using temperature sensors, an outdoor temperature sensor and a gas meter sensor. This model can predict the indoor temperature trends up to 90th percentile aggregate with an approximate error between 0.61°C and 1.50°C, especially in applications such as a boiler or a furnace actuation and for outdoor temperature forecasts.

with 'resolution', 'size' depending on the device on which such a proximity sensor will be mounted, and operating environment/conditions under which such a sensor will function." Such a sensor can have high reliability and long functional life due to absence of mechanical parts and lack of physical contact between the sensor and the object.

Reliability. Ensuring a robust design with maintenance-free packaging is very important. This is because sensors are that part of the device which mostly comes into contact with the world, and hence has to have the ability to withstand harsh environment without losing on performance. Sahil Khan, embedded systems consultant at Upheave Systems Pvt Ltd, adds that, "We have seen soil moisture sensors losing accuracy in saline agricultural soils. With huge requirements for the IoT, maintenance is a big no-no for a sensing node." A sensor must be able to transduce information reliably over and over again, and that too within reasonable accuracy.

Easy integration. For the developer's perspective, using ADC, employing filtering equations, extracting data and then calibrating the firmware is troublesome. This is why, to reduce design effort, pre-calibrated sensors which

support common interfaces like TWI/I²C or SWI, SPI, etc are preferable.

A common example is the use of NTC or PTC thermistors versus a DHT11 sensor to read temperature and humidity data. "DHT11 temperature and humidity sensor features a temperature and humidity sensor complex with a calibrated digital signal output. By using the exclusive digital signal acquisition technique and temperature and humidity sensing technology, it ensures high reliability and excellent long-term stability. This sensor includes a resistive-type humidity measurement component and an NTC temperature measurement component, and connects to a high-performance 8-bit microcontroller, offering excellent quality, fast response, anti-interference ability and

cost-effectiveness," explains Sahil.

He adds another example, the SP120S01RB sensor by National Semiconductor, which is used for measuring liquid level utilising a pressure sensor. Most importantly, this sensor is being used for sensing the quantity of water in drought-affected areas.

Software. There are also some sensors whose role mainly depends on the software. These analyse the environment by sensing some parameters and make use of a software to come to a decision. Fuzzy logic plays a vital role in such sensors. Chow Reddy explains, "Another case where sensor plays a small role but circuitry and software play a vital role is load cell for weighing applications. The load cell here is a wheatstone bridge to develop a differential potential at the stress applied on one of the wings of the bridge. The developed potential puts in few microvolts prone to get affected by noise due to external disturbances, vary due to vibrations created over the platform. ADC employed to read these parameters need to be properly selected." Chow Reddy further explains that software to reject the noise as well as average the weight data to produce a calibrated weight accurately is as

2013 Sensors Expo and Conference

Sensors Expo and Conference 2013 at Massachusetts proved to be an essential platform for the companies all around the world to provide an in-depth analysis on the use of sensors as an edge of Internet of Things (IoT), to present the stronger and faster MEMS sensor fusion, to provide an understanding of the cutting-edge sensing applications, remote monitoring, etc.